INCH-POUND

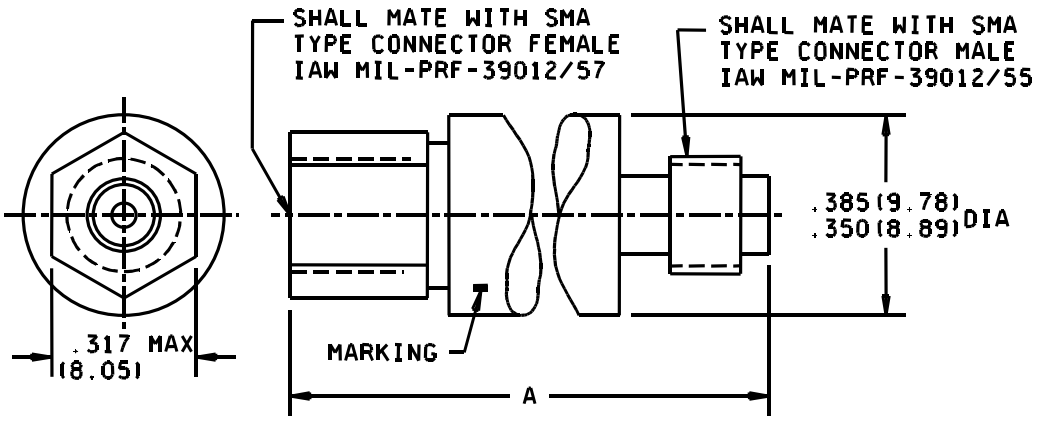
MIL-DTL-3933/14E
7 April 2011
SUPERSEDING
MIL-DTL-3933/14D
7 November 2002

DETAIL SPECIFICATION SHEET

ATTENUATORS, FIXED, SPACE LEVEL, NON SPACE LEVEL, COAXIAL LINE (SMA CONNECTORS), FREQUENCY RANGE: DC TO 12.4 GHz, CLASS IV, LOW POWER.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-3933.



NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Metric equivalents are in parentheses.

FIGURE 1. Dimensions and configuration.

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ENGINEERING DATA:

Operating frequency range: dc to 12.4 GHz.

Test frequency range for SWR and attenuation measurements, measured or recorded, when using Automatic Network Analyzer (ANA) or equivalent test equipment:

100 MHz (or lower) to 12.4 GHz.

Basic test frequency for sensitive measurements like connector repeatability, temperature sensitivity, power sensitivity and any other measurements where a 'delta' specification is imposed: 6.0 GHz

Operating temperature range: -55°C to +125°C.

Maximum weight: See table I.

REQUIREMENTS:

Dimensions and configuration: See figure 1.

Nominal impedance: 50 ohms.

Attenuation: See table I. Stability and sensitivity:

Maximum change in attenuation after:	up to 10 dB, inclusive (dB)	over 10 dB (dB/dB)
Temperature change or thermal shock	0.5	0.05
Vibration or shock	0.5	0.05
Moisture resistance or salt spray	0.5	0.05
Peak power	0.5	0.05

Frequency sensitivity, maximum: 0.4 dB/dB/GHz.
Temperature sensitivity of attenuation, maximum: 0.0006 dB/dB/°C.

Power: See table I.
Power sensitivity for full input power, maximum: 0.005 dB/dB/Watt.

VSWR: See table I.

Connector repeatability, maximum variation in attenuation: 0.04 dB.

Part or Identifying Number (PIN): M3933/14- (and dash number from table I with three device level designators are space level (T), screened (S) and non-screened (N) options).

Supersession data: See table II.

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TABLE I. Electrical characteristics.

Dash number		Attenuation dB		Maximum power input at 25°C		VSWR maximum 1/			Weight (oz) max	Dimension A (inches) maximum
		Nominal	Deviation DC to 12.4 GHz	AV (W) 3/ (continuously)	Peak 2/ (kW)	dc to 4.0 GHz	4.0 to 8.0 GHz	8.0 to 12.4 GHz		
01	N S T	3	±0.3	2	0.2	1.15:1	1.20:1	1.25:1	0.5	1.240 (31.50)
02	N S T	6	±0.3	2	0.2	1.15:1	1.20:1	1.25:1	0.5	1.240 (31.50)
03	N S T	10	±0.3	2	0.2	1.15:1	1.20:1	1.25:1	0.5	1.240 (31.50)
04	N S T	20	±0.5	2	0.2	1.15:1	1.20:1	1.25:1	0.5	1.240 (31.50)
05	N S T	15	±0.5	2	0.2	1.15:1	1.20:1	1.25:1	0.5	1.240 (31.50)
06	N S T	1	±0.3	2	0.2	1.15:1	1.20:1	1.25:1	0.5	1.240 (31.50)
07	N S T	2	±0.3	2	0.2	1.15:1	1.20:1	1.25:1	0.5	1.240 (31.50)
08	N S T	4	±0.3	2	0.2	1.15:1	1.20:1	1.25:1	0.5	1.240 (31.50)
09	N S T	5	±0.3	2	0.2	1.15:1	1.20:1	1.25:1	0.5	1.240 (31.50)
10	N S T	7	±0.4	2	0.2	1.15:1	1.20:1	1.25:1	0.5	1.240 (31.50)
11	N S T	8	±0.4	2	0.2	1.15:1	1.20:1	1.25:1	0.5	1.240 (31.50)
12	N S T	9	±0.4	2	0.2	1.15:1	1.20:1	1.25:1	0.5	1.240 (31.50)
13	N S T	30	±0.9	2	0.2	1.15:1	1.20:1	1.25:1	0.8	1.850 (46.99)

See footnotes at end of table.

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TABLE I. Electrical characteristics - Continued.

Dash number		Attenuation dB		Maximum power input at 25°C		VSWR maximum 1/			Weight (oz) max	Dimension A (inches) maximum
		Nominal	Deviation DC to 12.4 GHz	AV (W) 3/ (continuously)	Peak 2/ (kW)	dc to 4.0 GHz	4.0 to 8.0 GHz	8.0 to 12.4 GHz		
14	N S T	40	±0.9	2	0.2	1.15:1	1.20:1	1.25:1	0.8	1.850 (46.99)
15	N S T	60	±2.0	2	0.2	1.15:1	1.20:1	1.25:1	0.8	1.850 (46.99)
16	N S T	50	±2.0	2	0.2	1.15:1	1.20:1	1.25:1	0.8	1.850 (46.99)
17	N S T	28	±0.9	2	0.2	1.15:1	1.20:1	1.25:1	0.7	1.850 (46.99)
18	N S T	16	±0.5	2	0.2	1.15:1	1.20:1	1.25:1	0.5	1.240 (31.50)
19	N S T	14	±0.5	2	0.2	1.15:1	1.20:1	1.25:1	0.5	1.240 (31.50)
20	N S T	13	±0.5	2	0.2	1.15:1	1.20:1	1.25:1	0.5	1.240 (31.50)
21	N S T	12	±0.5	2	0.2	1.15:1	1.20:1	1.25:1	0.5	1.240 (31.50)
22	N S T	11	±0.5	2	0.2	1.15:1	1.20:1	1.25:1	0.5	1.240 (31.50)
23	N S T	1.5	±0.3	2	0.2	1.15:1	1.20:1	1.25:1	0.5	1.240 (31.50)
24	N S T	31	±0.9	2	0.2	1.15:1	1.20:1	1.25:1	0.8	1.850 (46.99)

1/ VSWR value is for both ends.
2/ Peak power for a duty cycle of 5×10^{-4}, maximum pulse duration 5 microseconds.
3/ Power input is derated linearly from +25°C to 0.5 watts at +125°C.

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TABLE II. Supersession data.

PIN M3933/14-	Supersedes M3933/15-
01	01
02	02
03	03
04	04
13	05
14	06
13	07
14	08

Referenced documents. In addition to MIL-DTL-3933 this specification sheet references the following documents:
MIL-PRF-39012/55
MIL-PRF-39012/57

Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodians:
Army - CR
Navy - EC
Air Force - 85
DLA - CC

Review activities:
Army - MI
Navy - AS, MC
Air Force - 99

Preparing activity:
DLA - CC

(Project 5985-2010-013)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.daps.dla.mil.

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